

Strathmore
UNIVERSITY

PRACTICAL EXERCISE 3

Group: CTRL + ALT + ELITE

Group Members

166142 – Njihia Muranga
152383 – Phillip Leo Jnr
168328 – Abucheli Ronah
168333 – Danni Podho
168971 – Philip Gakuo

EXPERIMENT 1

1. Pick one resistor, take a picture of it and determine its resistance value based on the colour code. Write down the resistance value.



**Red (2)
Red (2)
Orange (x1000)
Gold ($\pm 5\%$)**

Total Value: 22 k Ω

2. Using a multimeter, measure the resistance value emitted by the resistor. Take a picture of this setup and write down the measured resistance.



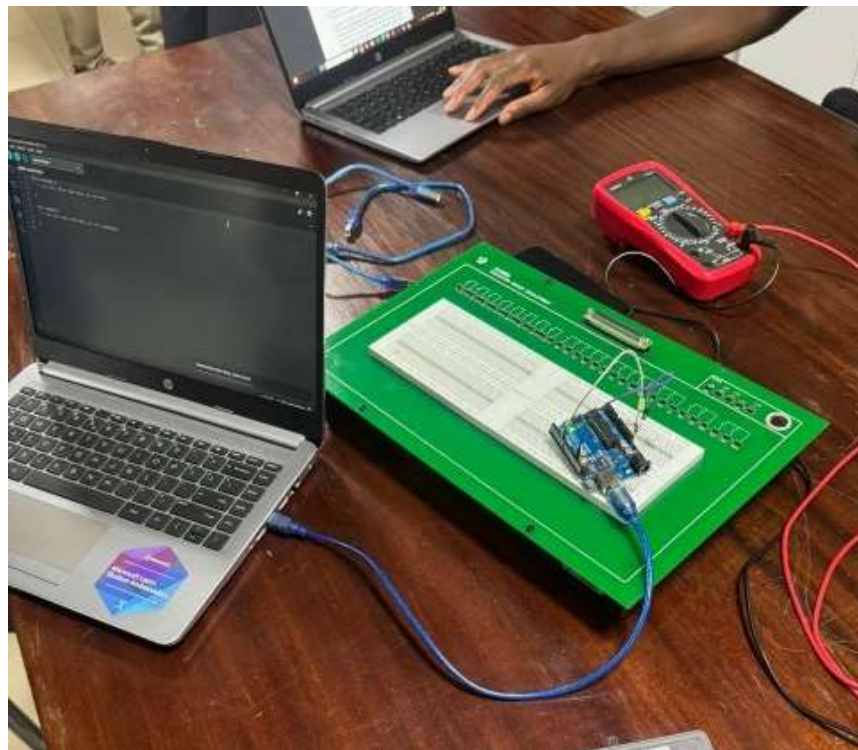
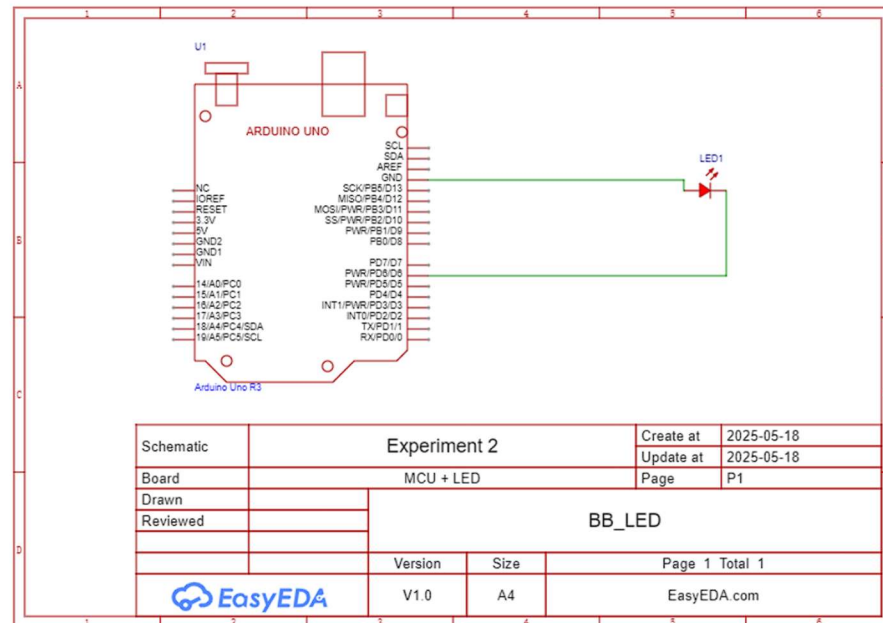
Experiment Reflection: Where does the difference in the resistance values arise from?

The difference comes from the tolerance value indicated by the gold band on the resistor. $\pm 5\%$ means that the expected value should be within 20.9 to 23.1 k Ω .

EXPERIMENT 2

1. Using the breadboard, microcontroller, and LED, recreate the following schematic.

Take a picture of the physical implementation.



2. In Arduino IDE, deploy the Blink program which is usually inbuilt in Arduino and upload to your board.

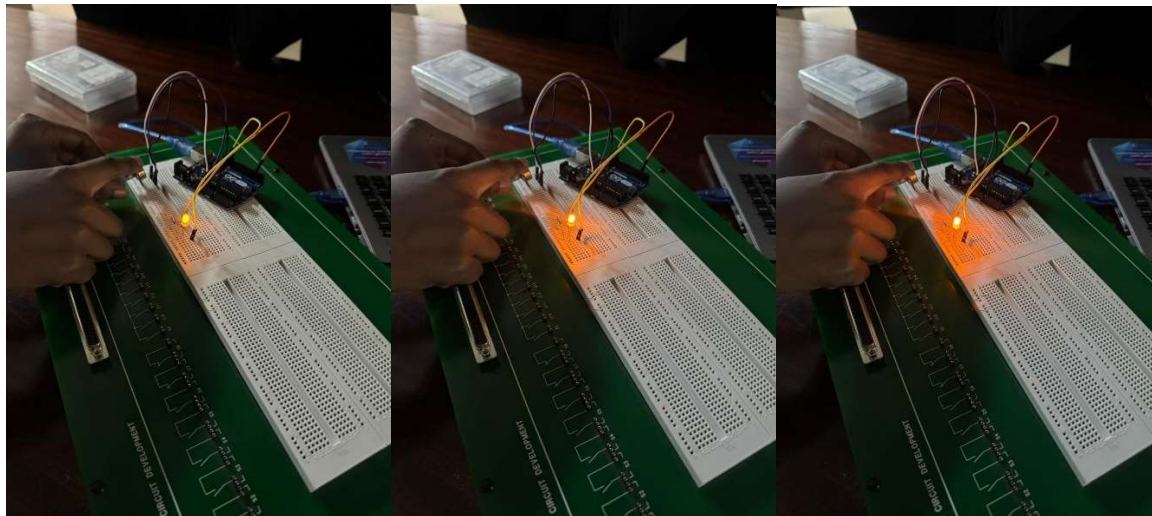
3. Thereafter, use the multimeter to determine the voltage passing through the circuit.

Write down the value registered.



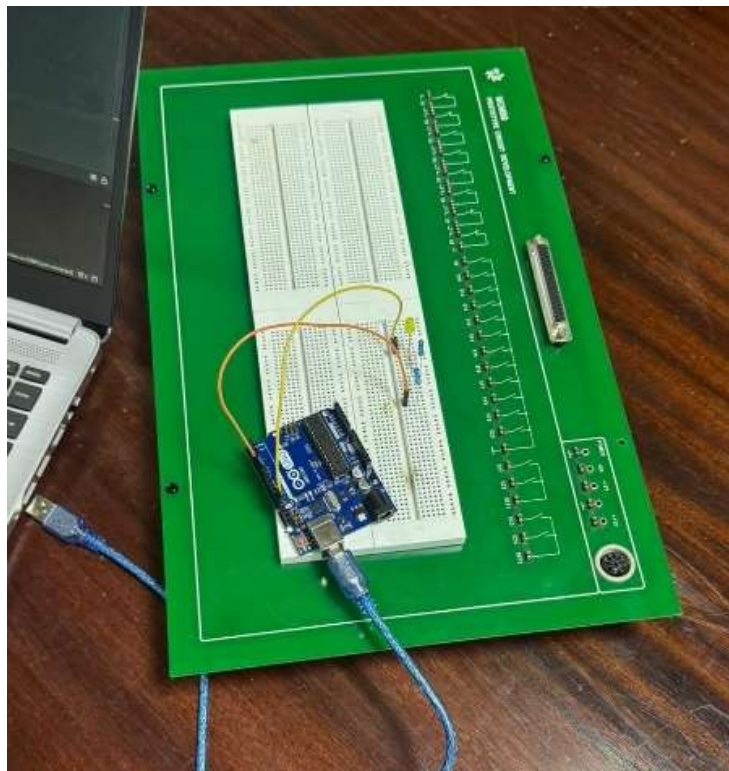
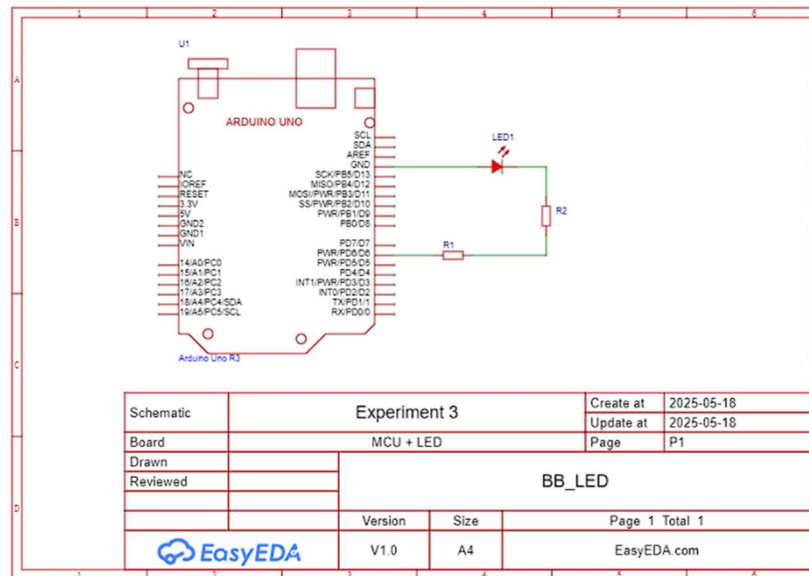
Observed Value:
2.21V

4. Reconfigure the circuit you created in question 1 by adding a potentiometer to simulate Pulse Width Modulation (PWM) on the LED. Take 3 images of your circuit with the LED at 25% brightness, 50% brightness and 100% brightness.

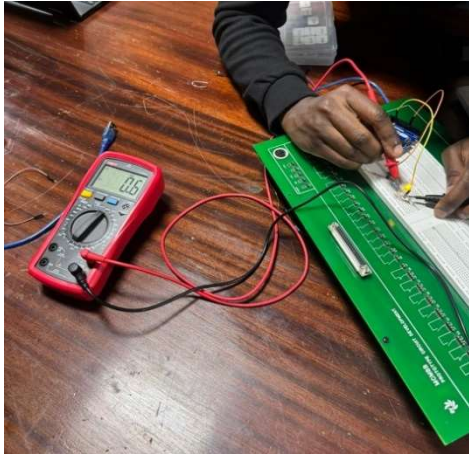


EXPERIMENT 3

1. Using the breadboard, microcontroller, both resistors and LED, recreate the following schematic of resistors connected in series. Take a picture of the physical implementation.



2. In Arduino IDE, re-deploy the Blink program and upload to your board.
 3. Thereafter, use the multimeter to determine the voltage passing through the circuit.
- Write down the value registered at each resistor.



LED: 0.6 mv

Resistor 1 (47Ω): 0.3 mv

Resistor 2 (47Ω): 0.3 mv

Experiment Reflection: Use Ohm's Law to estimate the current passed at each resistor.

(Since we used 2 of the same resistor)

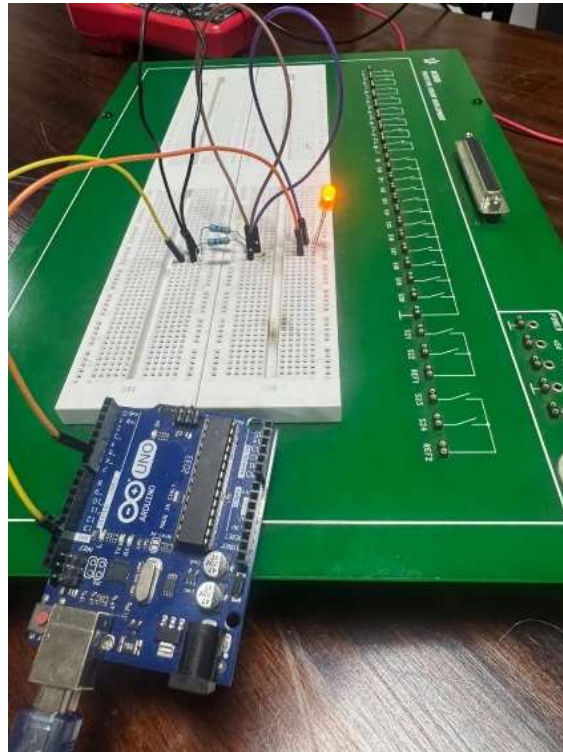
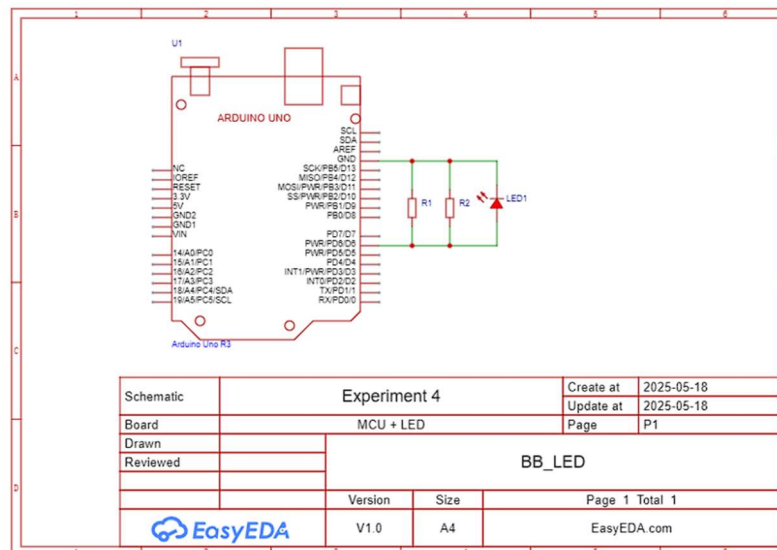
$$\mathbf{V = IR}$$

$$\mathbf{I = V/R}$$

$$\mathbf{= 0.0003/47 = 6.38 \mu A}$$

EXPERIMENT 4

1. Using the breadboard, microcontroller, both resistors and LED, recreate the following schematic of resistors connected in parallel. Take a picture of the physical implementation.



2. In Arduino IDE, re-deploy the Blink program and upload to your board.
3. Thereafter, use the multimeter to determine the voltage passing through the circuit.

Write down the value registered at each resistor.



Resistor 1 (47Ω): 1.1 V

Resistor 2 (47Ω): 1.1 V

Experiment Reflection: Use Ohm's Law to estimate the current passed at each resistor.

(Since we used 2 of the same resistor)

$$I = V/R$$

$$= 1.1/47 = 0.023 \text{ A}$$